Glaze Search · How to use

A friendly guide with **real examples**. Every photo, number and recipe below comes from the app itself, run on 2026-11-05. The deeper parts are in yellow 🐇 **rabbit hole** boxes: skip them on a first read.

1 What it does · 2 Start it · 3 The six pages · 4 Example A: test tiles · 5 Examples B–D: vase, bowl, mug · 6 Words only · 7 A formula · 8 All answers · 9 What the numbers mean · 10 Photo + words + formula · 11 Tips · 12 Where things are

1 · What the app does

You show it a pot. It tells you **what it sees** and **finds glaze recipes** that could make a similar glaze.

📷 What is this pot?

From a photo: the glaze colour, the shape and the surface, each with how sure the model is.

🧪 Which recipe makes it?

The closest of 7,063 real recipes from Glazy, from words, a colour, the photo or a chemical formula.

🏺 What looks like it?

The most similar pots among 12,309 objects cut out of Glazy photos, with ✓ / ✗ for shape, glaze and colour.

Everything runs on your own computer; only the recipe photos are loaded from Glazy's website (internet needed for them).

2 · Start the app

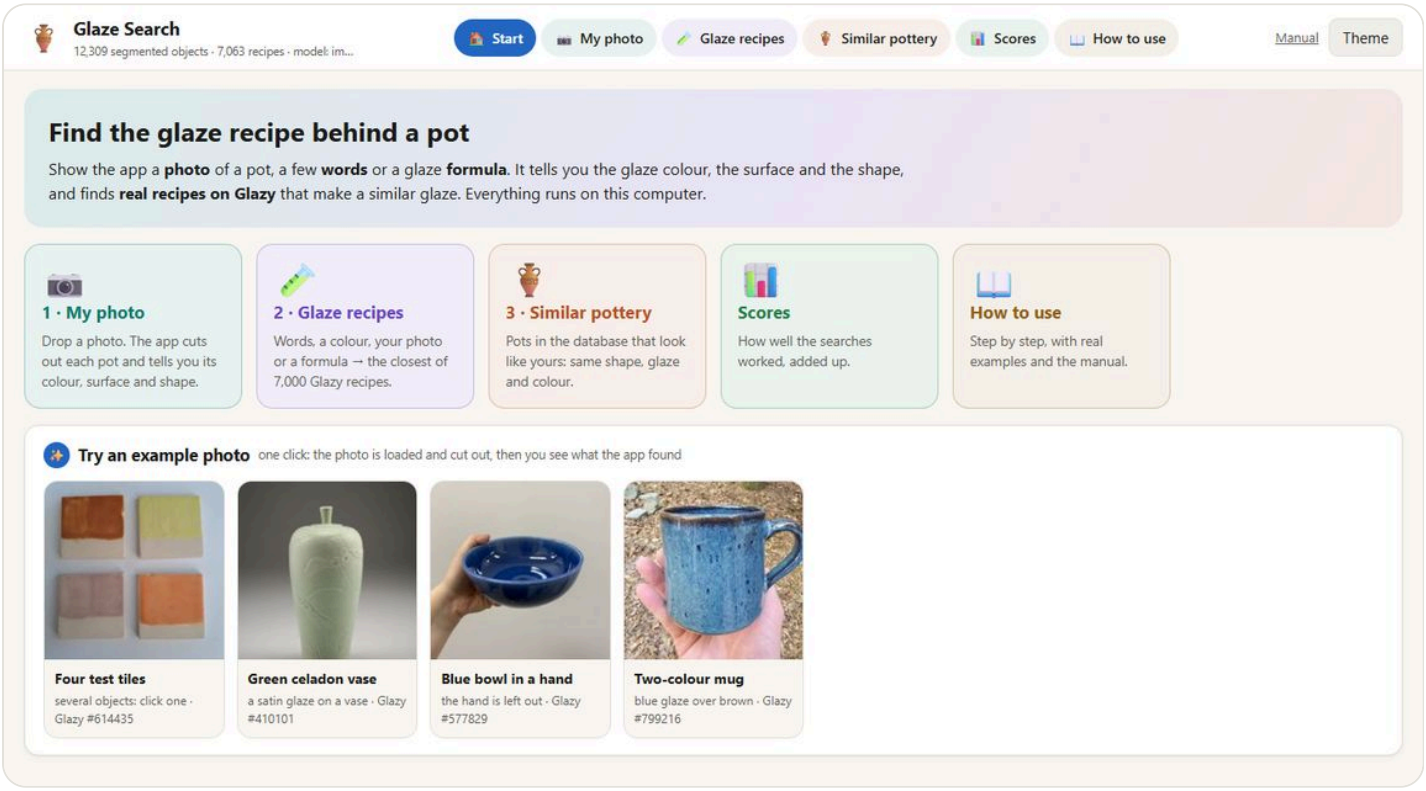
1. Double-click **RUN_GLAZE_SEARCH.bat** (in the project folder). A black window opens: **keep it open** while you use the app; closing it stops the app.
2. Wait about 30–60 seconds until the window says *Glaze recipe model ready* and *Smart search model ready*. The page opens in your browser at **http://127.0.0.1:8770**.
3. Use the **coloured buttons at the top** to move between pages. Your photo and choices stay when you switch.

First time on a new PC: SETUP.bat → DOWNLOAD_DATA.bat → CHECK_DATA.bat → INSTALL_GLAZE_MODEL.bat (with the team's model ZIP), then RUN_GLAZE_SEARCH.bat.

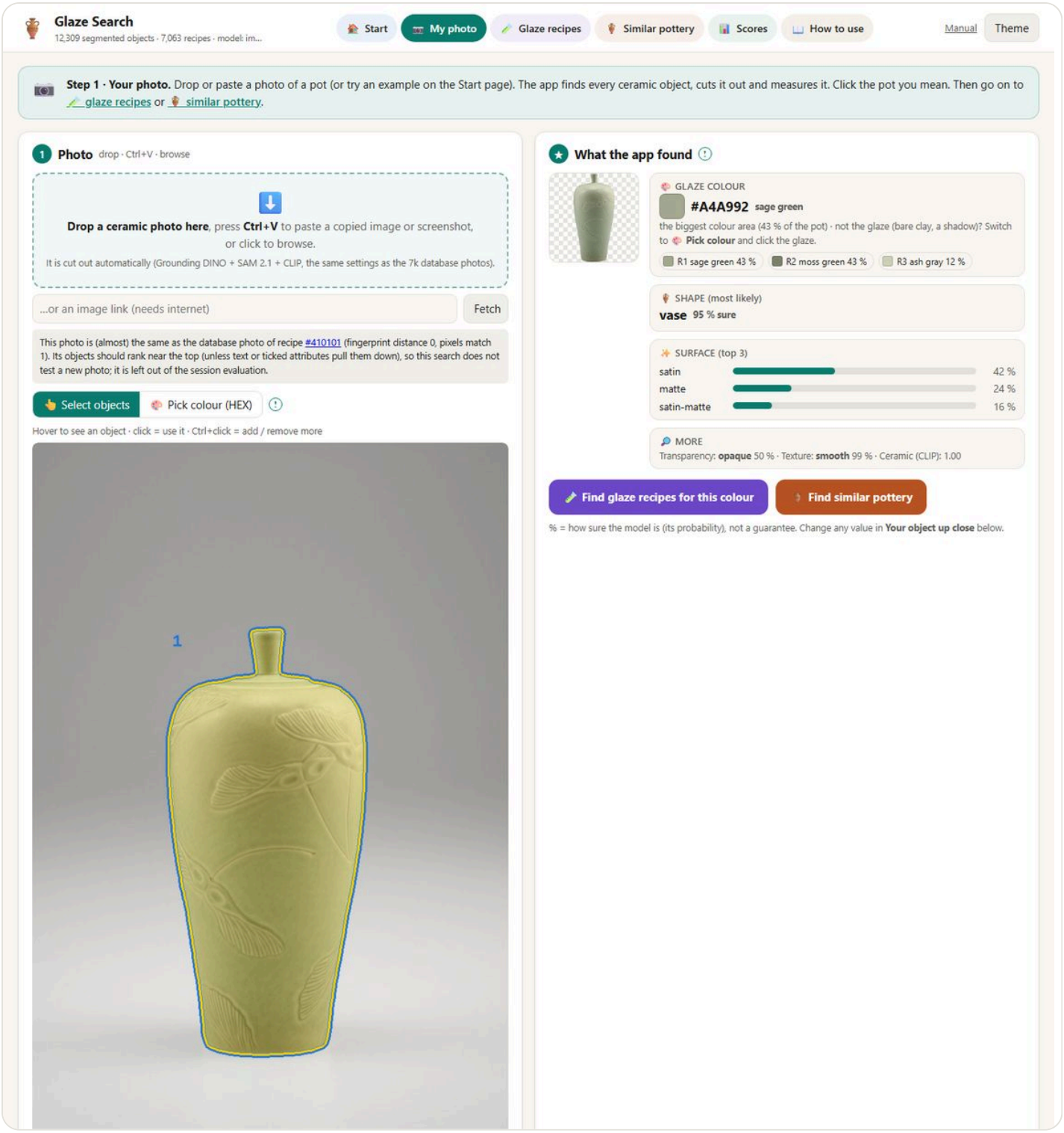
3 · The six pages

Each page has its own colour, and each one does one job.

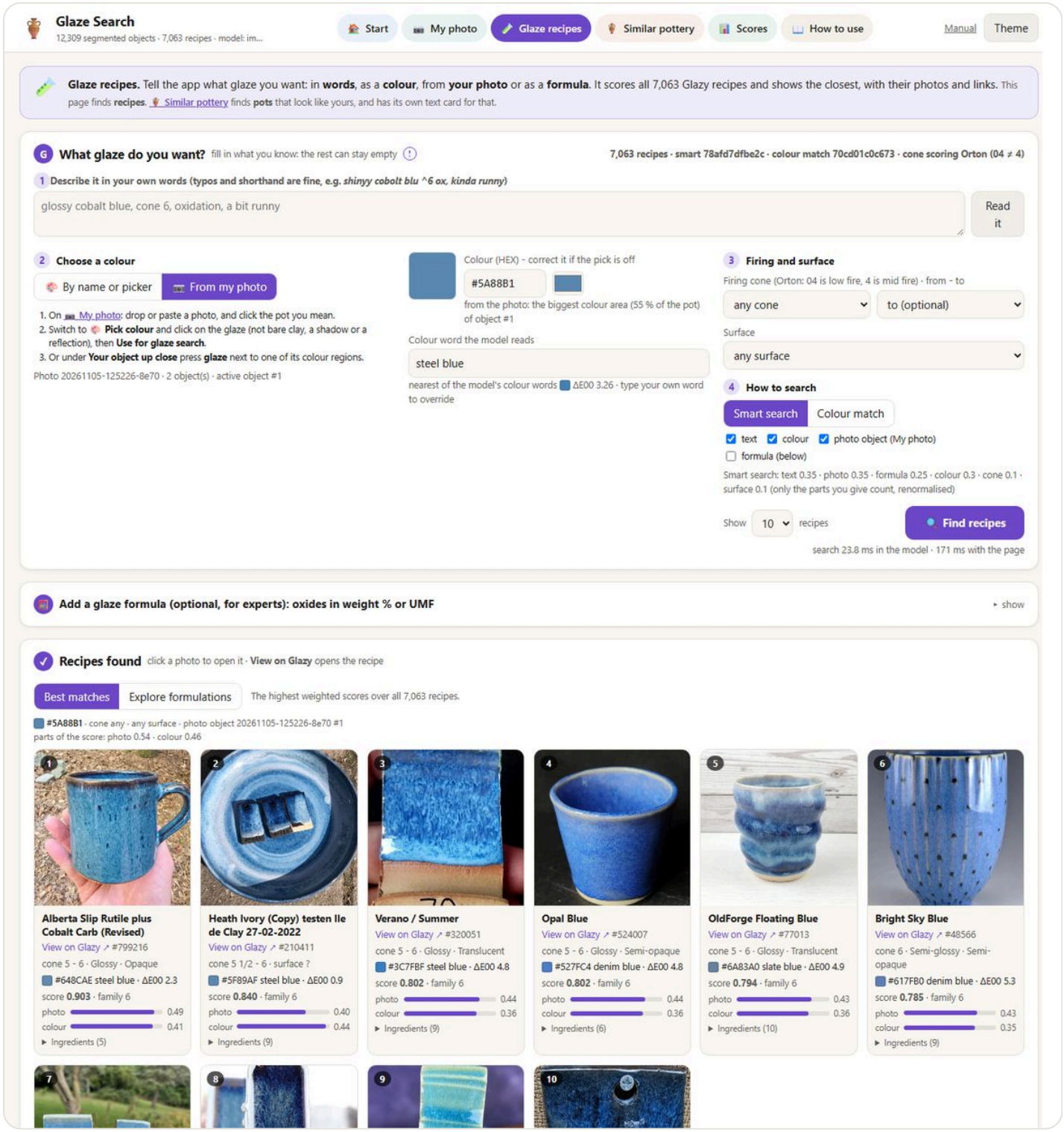
Button	What you do there	What you get
🏠 **Start**	Pick a page, or click an example photo	A one-click way in
📷 **My photo**	Drop / paste / browse a photo, click the pot, pick its glaze colour	**What the app found**: colour, shape, surface, more
🧪 **Glaze recipes**	Words, a colour, the photo, cone, surface, a formula; choose 3–20 results	Recipes with photos, **View on Glazy** links and score parts
🏺 **Similar pottery**	Choose what matters (Balanced, Shape, Glaze, Colour or your own sliders)	Similar pots with ✓ / ✗ and this search's scores
📊 **Scores**	Just look	All pottery searches added up (Hit Rate, Precision, nDCG)
📖 **How to use**	Read, or try an example	Quick start, example photos and this manual



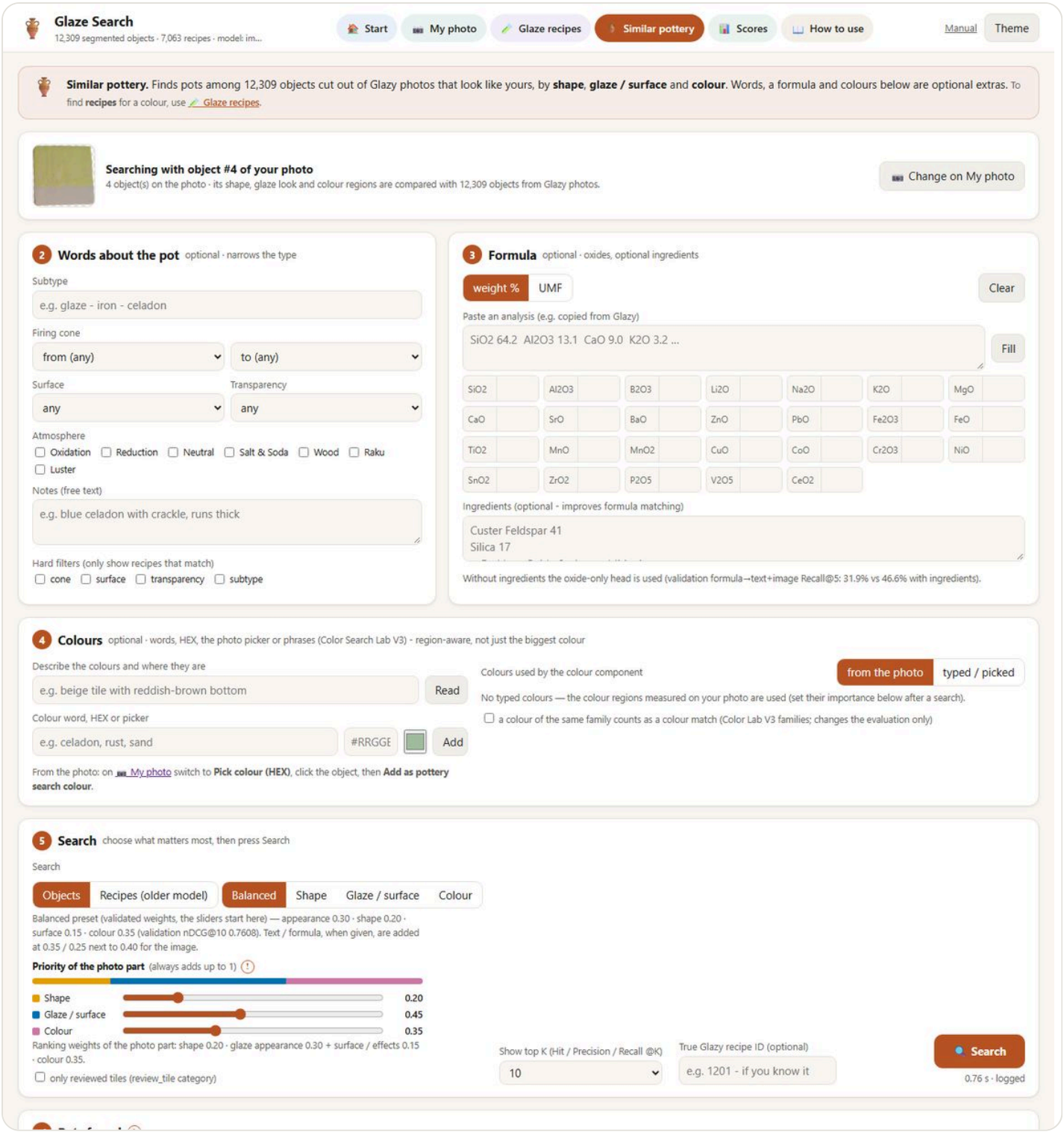
🏠 Start: the five big buttons and the example photos.



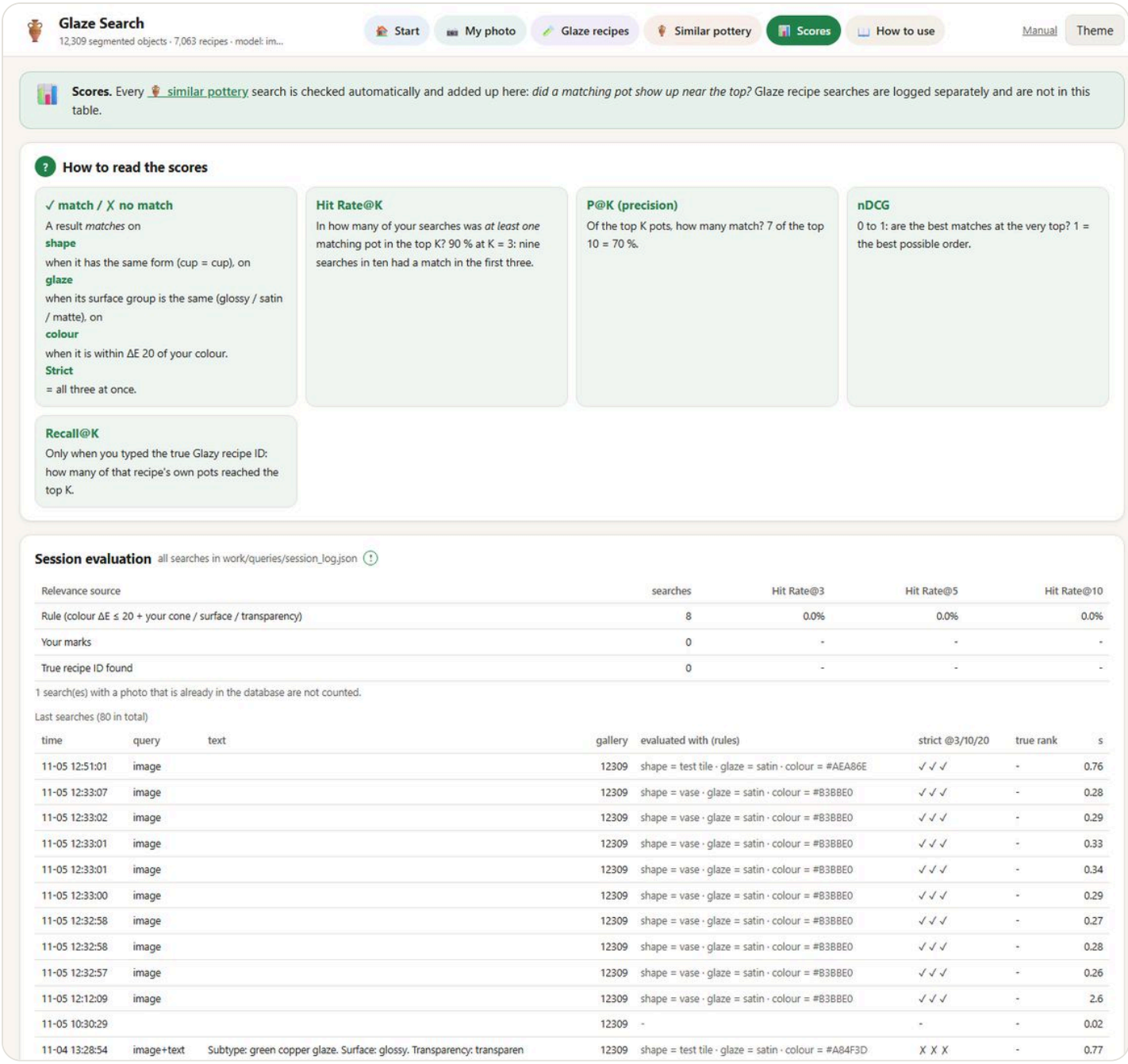
📷 My photo with the celadon vase (example B): the photo on the left, **What the app found** on the right, and below it **Your object up close** with all six features and their top 3 candidates.



Glaze recipes: four numbered steps (words, colour, firing and surface, how to search), the **Show 3 / 5 / 10 / 15 / 20** choice, then the recipes.



Similar pottery: the photo strip on top, optional words / formula / colours, the search settings, then the pots with ✓ / ✗.



📊 Scores: plain-words help on top, then the session table.

4 · Example A: four test tiles, step by step



Example A

Four test tiles: the yellow one

Photo from Glazy: Satin matte base glaze · recipe #614435 ↗. It is one of the 7,063 database photos, so the app says so and leaves this search out of the session scores.

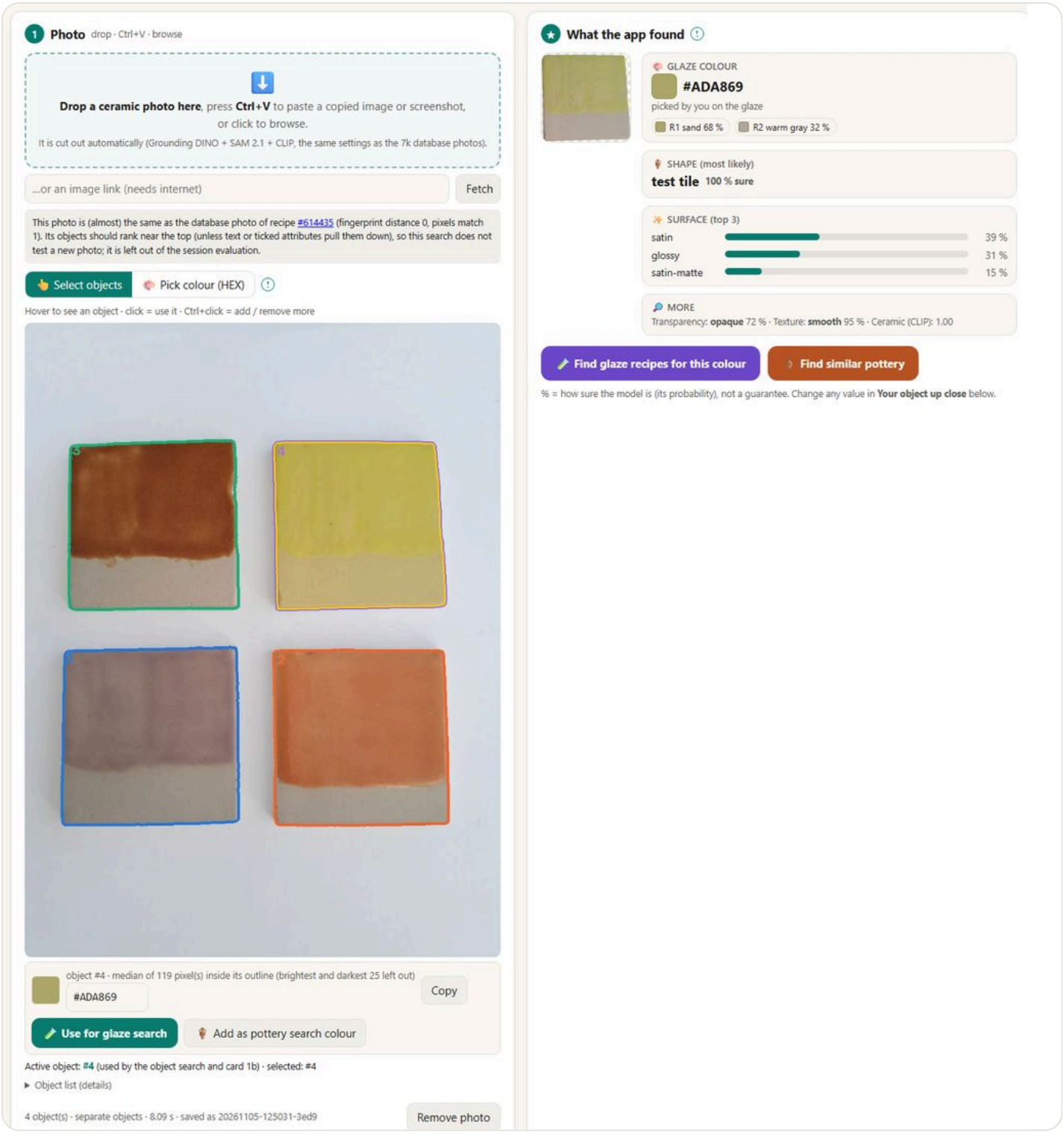
Step 1 · 📷 My photo: find the pots and pick the glaze

What you do

Click the example (or drop the photo). Click the **yellow tile**. Switch to **🎨 Pick colour** and click on the yellow glaze, not on the bare clay at the bottom.

What you see

4 objects, each with a coloured outline; the active one is yellow. The picked colour is **#ADA869** (the middle value of 119 pixels inside the tile).





GLAZE COLOUR

#ADA869

picked by you on the glaze

R1 sand 68 % · R2 warm gray 32 %

SHAPE (most likely)

test tile 100 % sure

SURFACE (top 3)

satin	39 %
glossy	31 %
satin-matte	15 %

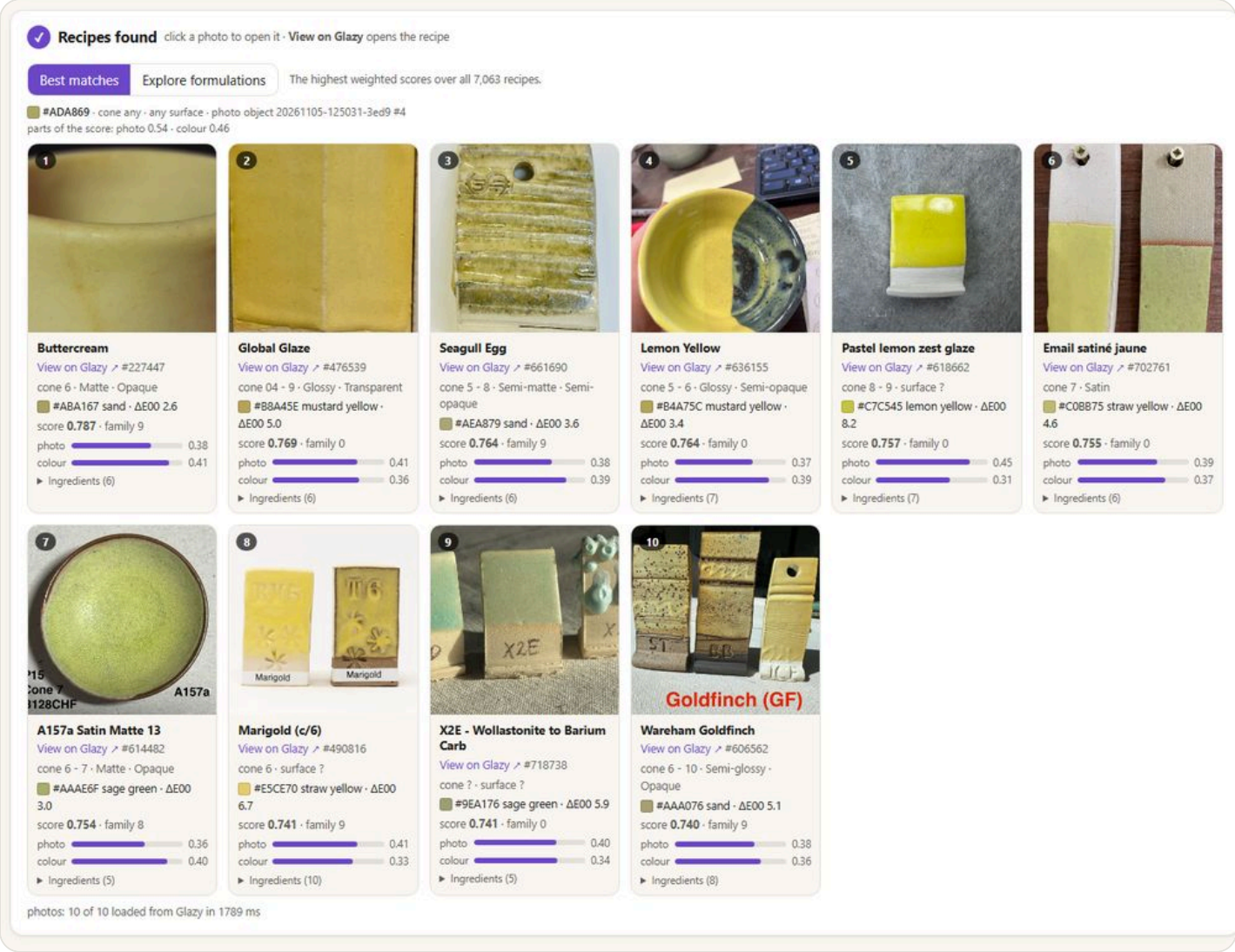
MORE

transparency: **opaque** 72 % · texture: **smooth** 95 % · ceramic (CLIP) 1.00 · 4 object(s) on the photo

> **Rabbit hole: where do these answers come from?**
>
> - **Cutting out:** Grounding DINO draws a box around each “ceramic object”, SAM 2.1 traces its exact outline, and CLIP checks it is ceramic (1.00 here; below 0.35 an object is not used by default). That is why a hand or a table is left out (see example C).
> - **Colour regions:** inside the outline, pixels with similar colour are grouped. This tile has two regions: R1 sand 68 % (the glaze) and R2 warm gray 32 % (the bare clay). Picking with 🎨 makes sure the **glaze** is used.
> - **Shape and surface:** Grounding DINO looks for phrases such as “flat rectangular tile” or “mirror-like reflection”; a small trained layer turns that into a probability. Shape test tile 100 % is almost certain. The surface is not: satin 39 %, glossy 31 %, satin-matte 15 %. Glazy says **satin-matte**, which is only the model's third choice. Surfaces are hard to see in a photo (on 240 test objects the first surface guess was right 45 % of the time, the first shape guess 84 %).

Step 2 · 🧪 Find glaze recipes for this colour

Press 🧪 **Find glaze recipes for this colour**. The app opens the recipes page and searches with the **photo** and the **picked colour**.



The top of the result list (Smart search, photo + colour).



Buttercream

View on Glazy ↗ #227447

cone 6 · Matte

#ABA167 sand · ΔE00 2.6

score **0.787** · family 9

photo 0.38

colour 0.41



Global Glaze

View on Glazy ↗ #476539

cone 04 - 9 · Glossy

#B8A45E mustard yellow · ΔE00 5.0

score **0.769** · family 0

photo 0.41

colour 0.36



Seagull Egg

View on Glazy ↗ #661690

cone 5 - 8 · Semi-matte

#AEA879 sand · ΔE00 3.6

score **0.764** · family 9

photo 0.38

colour 0.38

🐇 **Rabbit hole: how is the score 0.787 of "Buttercream" made?**

Only the parts you gave count. Here that is the **photo** (weight 0.35) and the **colour** (0.30). Their weights are scaled to add up to 1: photo 0.35 / 0.65 = **0.54**, colour 0.30 / 0.65 = **0.46**.

```
colour part = exp(−ΔE00 / 20) = exp(−2.58 / 20) = 0.879
```

```
score = 0.5385 × 0.709 (photo) + 0.4615 × 0.879 (colour) = 0.787
```

The bars on each card show the same thing: the bar length is the part's own score (0–1), the number is its share of the total.

Why is the tile's own recipe not first? Glazy's photo of "Satin matte base glaze" shows **four** colours. The photo part compares your yellow tile with the whole recipe photo, so other, all-yellow recipes are closer. Its rank here: not in the top 10. With a one-glaze photo the own recipe does come first (examples C and D).

Step 3 · Add words: "satin matte, cone 6"

Type the words in **1 · Describe it** and press Find recipes again. The app reads them: **cone 6** and **Satin-matte**. Now five parts count, so the weights are scaled again: words 0.29 · photo 0.29 · colour 0.25 · cone 0.08 · surface 0.08.



Base for inclusion stains (Neph Sye, 3134)

View on Glazy ↗ #643772

cone 6 · Satin-matte

#B8B48A sand · ΔE00 6.2

score **0.763** · family 3

words	0.21
photo	0.21
colour	0.18
cone	0.08
surface	0.08

✓ cone ✓ surface



MagMattLinerVI

View on Glazy ↗ #582121

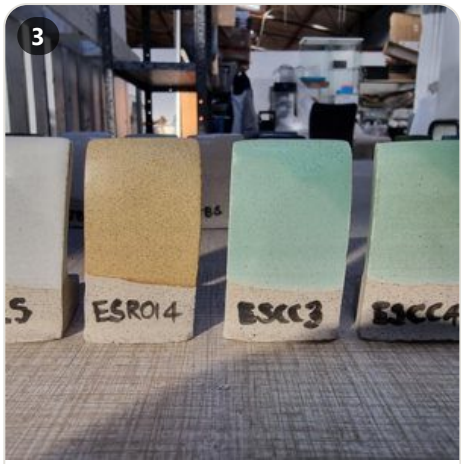
cone 6 - 7 · Satin-matte

#C9C0AA oatmeal · ΔE00 9.6

score **0.735** · family 3

words	0.21
photo	0.20
colour	0.15
cone	0.08
surface	0.08

✓ cone ✓ surface



Eggshell Satin (Copy) using Frit 3134

View on Glazy ↗ #402586

cone 5 - 7 · Satin-matte

#B5AFA8 warm gray · ΔE00 11.0

score **0.727** · family 3

words	0.22
photo	0.19
colour	0.14
cone	0.08
surface	0.08

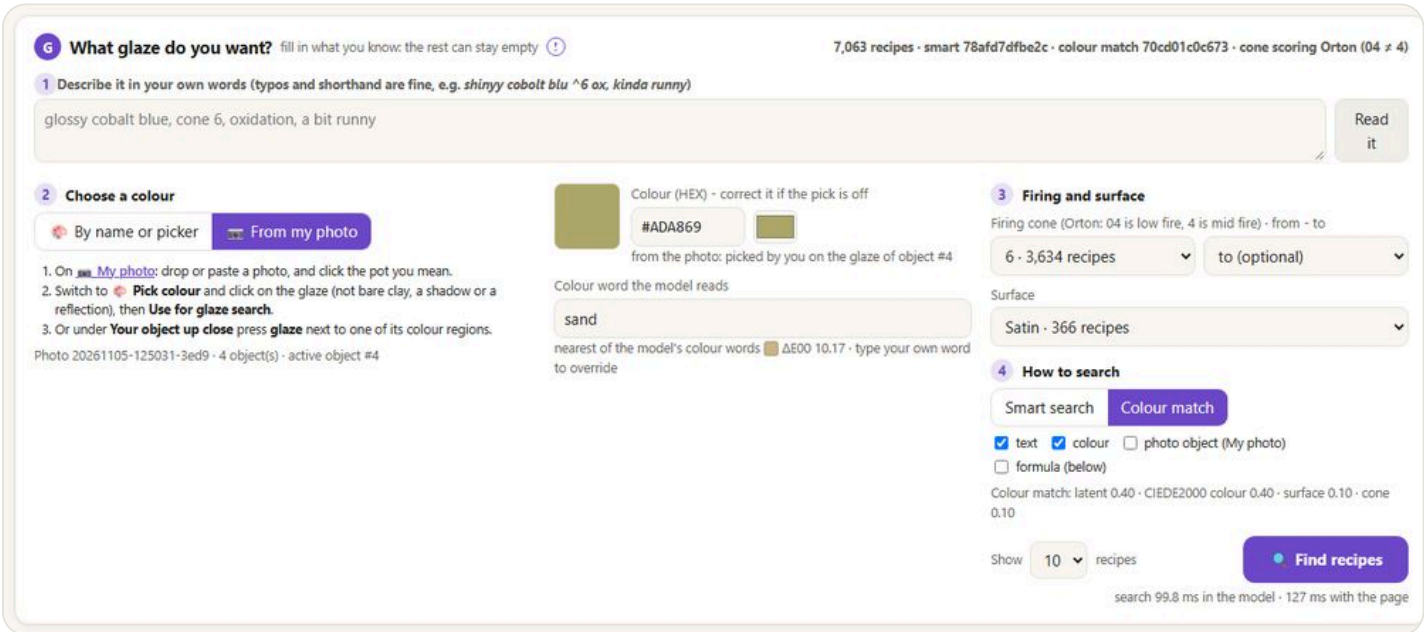
✓ cone ✓ surface

What changed: all three are now ✓ cone ✓ surface satin-matte glazes for cone 6. “Buttercream” drops to rank 5: its surface is Matte, not Satin-matte, so it gets 0 for surface (✗ surface), even though its colour is closest.

Step 4 · The team's Colour match model (colour + cone + surface)

Switch **4 · How to search** to **Colour match**, choose cone 6 and surface Satin. This model reads the words

`color: sand, Cone 6, Satin, unknown, unknown` and compares them with each recipe's **chemistry**.





MC6G HCSM2 - Beth's Test1 (Ru 7.2/ Cu 3.2/ Borate 10)

View on Glazy ↗ #643412

cone 6 · Satin

#B8B178 sand · ΔE00 3.2

score **0.763** · family 4

learned match		0.22
colour		0.34
surface		0.10
cone		0.10



Buttercream

View on Glazy ↗ #227447

cone 6 · Matte

#ABA167 sand · ΔE00 2.6

score **0.712** · family 5

learned match		0.26
colour		0.35
surface		0.00
cone		0.10



Yoko's Leaf Green

View on Glazy ↗ #410101

cone 5 - 6 · Satin

#929880 green gray · ΔE00 12.0

score **0.709** · family 4

learned match		0.29
colour		0.22
surface		0.10
cone		0.10

🐇 Rabbit hole: a match that is not the same colour

Score = 0.40 learned match + 0.40 colour + 0.10 surface + 0.10 cone. For #1:

```
0.40 × 0.553 + 0.40 × 0.854 + 0.10 × 1 + 0.10 × 1 = 0.763
```

#3 "Yoko's Leaf Green" is green-gray: ΔE00 12.0, so its colour part is only 0.55 (✗ **not the same colour**). It still ranks high because its chemistry fits the words best (learned match 0.73) and cone and surface match. That is why the app shows ΔE00 on every card: **check it**. Below about 5 the colours look alike; above 10 they are different colours.

Step 5 · 🏺 Find similar pottery

Press 🏺 **Find similar pottery** (on the photo page) or 🔍 **Search** on the pottery page. Balanced preset: shape 0.20, glaze 0.45, colour 0.35.

Pots found

SEARCHED	MODE	RULE HIT@10	YOUR HIT@10
12,309 segmented objects	**Balanced** shape 0.05 · glaze appearance 0.07 · surface / effects 0.04 · colour regions 0.08 · attributes (ticked) 0.16	**n/a** needs a colour + cone, surface or transparency	**-** mark results ✓ / ✗ below

NOTE	PRIORITY USED
database photo same photo as recipe #614435 — not counted in the session evaluation	**shape 0.20 · glaze 0.45 · colour 0.35** the mode preset (validated)

Evaluation of this search (separate from the ranking: it never changes the order)

Criterion (rule)	Hit@10	Precision@10	Hit@3 / 5 / 10 / 15 / 20	matching in gallery	Recall@10
Shape (object form) same object form: test tile	✓ yes	100.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	9,555	0.1% *(max 0.1%)*
Glaze / surface same surface group: satin (your object: satin)	✓ yes	30.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	4,359	0.1% *(max 0.2%)*
Colour within ΔE2000 20 of #AEA86E	✓ yes	90.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	9,295	0.1% *(max 0.1%)*
Strict (all criteria) shape AND glaze AND colour	✓ yes	20.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	2,798	0.1% *(max 0.4%)*
Weighted (your priority) shape 0.20 + glaze / surface 0.45 + colour 0.35	nDCG@10 **0.700**		@3 0.76 · @5 0.77 · @10 0.70 · @15 0.75 · @20 0.76		

▸ Ticked attributes check (card 1b ticks; every ticked attribute separately)

1



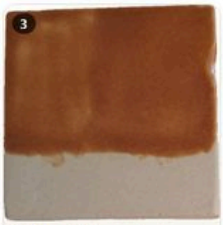
Satin matte base glaze 614435_4.jpg (photo 614435.jpg) · Glazy #614435
glaze - matte
cone 6 · satin-matte · semi-opaque
ΔE 0 ΔE 0
evaluation: ✓ shape ✓ glaze ✓ colour ✓ strict
weighted 1.00
ticked: ✓ colour ✓ object form: test tile (dino)
▸ Why this rank? · score 2.60
✓ relevant ✗ not

2



Satin matte base glaze 614435_2.jpg (photo 614435.jpg) · Glazy #614435
glaze - matte
cone 6 · satin-matte · semi-opaque
ΔE 15.5 ΔE 3.6
evaluation: ✓ shape ✗ glaze ✓ colour ✗ strict
weighted 0.55
ticked: ✓ colour ✓ object form: test tile (dino)
▸ Why this rank? · score 1.72
✓ relevant ✗ not

3



Satin matte base glaze 614435_3.jpg (photo 614435.jpg) · Glazy #614435
glaze - matte
cone 6 · satin-matte · semi-opaque
ΔE 15.4 ΔE 5.4
evaluation: ✓ shape ✗ glaze ✓ colour ✗ strict
weighted 0.55
ticked: ✓ colour ✓ object form: test tile (dino)
▸ Why this rank? · score 1.52
✓ relevant ✗ not

4



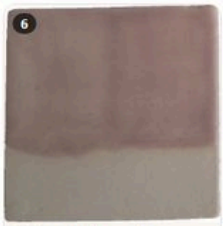
G2934 Matte Glaze Base for Cone 6 56097_1.jpg (photo 56097.jpg) · Glazy #56097
glaze - matte
cone 6 · surface ? · transparency ?
ΔE 18.9 ΔE 8.6
evaluation: ✓ shape ✓ glaze ✓ colour ✓ strict
weighted 1.00
ticked: ✓ colour ✓ object form: test tile (dino)
▸ Why this rank? · score 1.42
✓ relevant ✗ not

5



Stroke & Coat Base 69363_15.jpg (photo 69363.jpg) · Glazy #69363
glaze
cone 04–10 · surface ? · transparency ?
ΔE 15.8 ΔE 18
evaluation: ✓ shape ✗ glaze ✓ colour ✗ strict
weighted 0.55
ticked: ✓ colour ✓ object form: test tile (dino)
▸ Why this rank? · score 1.41
✓ relevant ✗ not

6



Satin matte base glaze 614435_1.jpg (photo 614435.jpg) · Glazy #614435
glaze - matte
cone 6 · satin-matte · semi-opaque
ΔE 25.1 ΔE 16.4
evaluation: ✓ shape ✓ glaze ✗ colour ✗ strict
weighted 0.65
ticked: ✗ colour ✓ object form: test tile (dino)
▸ Why this rank? · score 1.40
✓ relevant ✗ not

7



8



9



10





Satin matte base glaze
object
614435_4 · Glazy #614435 ↗
cone 6 · satin-matte
→ ΔE 0
✓ shape
✓ glaze
✓ colour
✓ strict



Satin matte base glaze
object
614435_2 · Glazy #614435 ↗
cone 6 · satin-matte
→ ΔE 15.5
✓ shape
✗ glaze
✓ colour
✗ strict



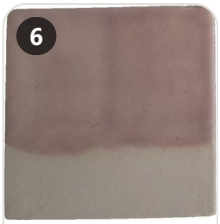
Satin matte base glaze
object
614435_3 · Glazy #614435 ↗
cone 6 · satin-matte
→ ΔE 15.4
✓ shape
✗ glaze
✓ colour
✗ strict



G2934 Matte Glaze Base for Cone 6
object 56097_1 · Glazy #56097 ↗
cone 6 · surface ?
→ ΔE 18.9
✓ shape
✓ glaze
✓ colour
✓ strict



Stroke & Coat Base
object
69363_15 · Glazy #69363 ↗
cone 04–10 · surface ?
→ ΔE 15.8
✓ shape
✗ glaze
✓ colour
✗ strict



Satin matte base glaze
object
614435_1 · Glazy #614435 ↗
cone 6 · satin-matte
→ ΔE 25.1
✓ shape
✓ glaze
✗ colour
✗ strict

Rabbit hole: reading ✓ and ✗, and this search's scores

Each pot is checked against simple rules (this check never changes the order):

criterion (rule)	Hit@3	Hit@10	P@10	matching pots in the database
shape same object form: test tile	✓	✓	100 %	9,555
glaze same surface group: satin (your object: satin)	✓	✓	30 %	4,359
colour within ΔE2000 20 of #AEA86E	✓	✓	90 %	9,295
strict shape AND glaze AND colour	✓	✓	20 %	2,798
weighted shape 0.20 + glaze 0.45 + colour 0.35	nDCG@10 **0.70** · nDCG@3 0.76			

- **#1** is the tile itself (the photo is in the database): ✓ everything.
- **#2 and #3** are the other tiles of the same photo: same shape and close colour, but their surface label is not satin, so ✗ glaze and ✗ strict.
- **#6** is the mauve tile: ΔE 25.1 is more than 20, so ✗ colour.
- **Hit@3** ✓ means at least one of the first 3 pots matches. **P@10 20 %** for strict means 2 of the first 10 pots match on all three. **nDCG@10 0.70**: 1 would be the best possible order.
- The colour rule uses the colour region the app thinks matters most (#AEA86E). Mark a region *important* under **Your object up close** to choose it yourself.

5 · Examples B, C and D: one pot each

Here the app chose everything itself: the pot (the active object) and the colour (the biggest colour area). Nothing was picked by hand.



Example B

A green celadon vase

Photo from Glazy: Yoko's Leaf Green · recipe #410101 ↗. It is one of the 7,063 database photos, so the app says so and leaves this search out of the session scores.



🎨 GLAZE COLOUR

#A4A992 sage green

the biggest colour area (43 % of the pot)

- R1 sage green 43 %
- R2 moss green 43 %
- R3 ash gray 12 %

🏺 SHAPE (most likely)

vase 95 % sure

✨ SURFACE (top 3)

satin	42 %
matte	24 %
satin-matte	16 %

🔎 MORE

transparency: **opaque** 50 % · texture: **smooth** 99 % · ceramic (CLIP) 1.00 · 1 object(s) on the photo

🧪 Top 3 recipes (photo + colour)

1



Jeannine's Sagebrush (SAC)

View on Glazy ↗ #690534

cone 6 · surface ?

#A1A492 sage green · ΔE00 2.5

score **0.802** · family 8

photo 0.39

colour 0.41

2



Yoko's Leaf Green

View on Glazy ↗ #410101

cone 5 - 6 · Satin

#929880 green gray · ΔE00 5.4

score **0.783** · family 8

photo 0.43

colour 0.35

3



Vert Crackle (ES)

View on Glazy ↗ #699504

cone 7 · Glossy

#47876D emerald green · ΔE00 2.5

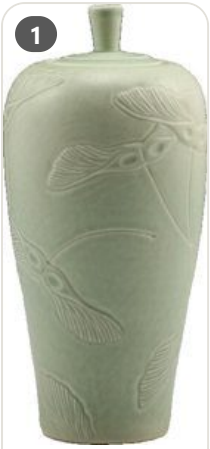
score **0.780** · family 8

photo 0.37

colour 0.41

Similar pottery (Balanced)

1



Yoko's Leaf Green
object
410101_1 · Glazy #410101 ↗
cone 5–6 · satin
→ ΔE 0
✓ shape
✓ glaze
✓ colour
✓ strict

2



Ice floe celadon Sr no Zn
object
146821_7 · Glazy #146821 ↗
cone 9–10 · surface ?
→ ΔE 9.7
✗ shape
✓ glaze
✓ colour
✗ strict

3



Yingqing (Qingbai) with Glaze Stone
object
738755_1 · Glazy #738755 ↗
cone 10 · glossy
→ ΔE 11.1
✓ shape
✗ glaze
✓ colour
✗ strict

4



Ice floe celadon Sr no Zn
object
146821_17 · Glazy #146821 ↗
cone 9–10 · surface ?
→ ΔE 9.8
✗ shape
✓ glaze
✓ colour
✗ strict

5



透明纹片
object
680771_2 · Glazy #680771 ↗
cone 7–9 · surface ?
→ ΔE 16.1
✗ shape
✗ glaze
✓ colour
✗ strict

6



Cone 6 Satin Mint (bone ash variation)
object
787708_1 · Glazy #787708 ↗
cone 6 · matte
→ ΔE 6.9
✗ shape
✓ glaze
✓ colour
✗ strict

Matched: the shape (vase) and the surface (satin, as on Glazy). The recipe's own entry is **#2**. #1 is a sage glaze whose colour is even closer (ΔE00 2.5).

Similar pottery: only 309 of the 12,309 database objects are vases, so after the vase itself most similar pots are celadons of another shape: ✗ shape, ✓ colour. Shape P@10 is 20 %. Choose the **Shape** preset if the shape matters most.



Example C

A blue bowl held in a hand

Photo from Glazy: Storm Blue - Blue Room Ingredients · recipe #577829 ↗. It is one of the 7,063 database photos, so the app says so and leaves this search out of the session scores.



🎨 GLAZE COLOUR

#1E345B navy blue

the biggest colour area (38 % of the pot)

- R1 navy blue 38 %
- R2 denim blue 14 %
- R3 blue black 11 %
- R4 navy blue 7 %
- R5 navy blue 6 %

🏺 SHAPE (most likely)

bowl 97 % sure

✨ SURFACE (top 3)

glossy 95 %

🔎 MORE

transparency: **opaque** 50 % · texture: **smooth** 100 % · ceramic (CLIP) 1.00 · 1 object(s) on the photo

🧪 Top 3 recipes (photo + colour)

1



Storm Blue - Blue Room Ingredients

View on Glazy ↗ #577829

cone 6 - 8 · Glossy

#1F3153 navy blue · ΔE00 1.7

score **0.913** · family 6

photo 0.49

colour 0.42

2



MG2 1Co 1Cu

View on Glazy ↗ #44644

cone 6 · Semi-glossy

#1F2D52 navy blue · ΔE00 3.0

score **0.816** · family 6

photo 0.42

colour 0.40

3



12 - Blue haze based on kitten's clear adapt with Gillespie

View on Glazy ↗ #673995

cone 6 · surface ?

#1E3557 navy blue · ΔE00 1.5

score **0.814** · family 6

photo 0.38

colour 0.43

Similar pottery (Balanced)

1



Storm Blue - Blue Room Ingredients
object
577829_1 · Glazy #577829 ↗
cone 6–8 · glossy
→ ΔE 1.8
✓ shape
✓ glaze
✓ colour
✓ strict

2



Navy Blue
object
31207_1 · Glazy #31207 ↗
cone 6 · glossy
→ ΔE 6.1
✓ shape
✓ glaze
✓ colour
✓ strict

3



Midnight Blue
object
30786_3 · Glazy #30786 ↗
cone 6 · surface ?
→ ΔE 5.8
✓ shape
✓ glaze
✓ colour
✓ strict

4



Midnight Blue
object
30786_2 · Glazy #30786 ↗
cone 6 · surface ?
→ ΔE 11.9
✓ shape
✓ glaze
✓ colour
✓ strict

5



Midnight Blue
object
30786_4 · Glazy #30786 ↗
cone 6 · surface ?
→ ΔE 12.2
✓ shape
✓ glaze
✓ colour
✓ strict

6



Midnight Blue
object
30786_6 · Glazy #30786 ↗
cone 6 · surface ?
→ ΔE 12.5
✓ shape
✓ glaze
✓ colour
✓ strict

The hand is left out: only the bowl is cut out (1 object, ceramic 1.00). The bowl has 9 colour regions (light on the rim, dark inside); the biggest one is used.

Matched: its own recipe is **#1** with score 0.913 (ΔE00 1.7). Similar pottery: nDCG@10 0.96, strict P@10 70 %.



Example D

A two-colour mug

Photo from Glazy: Alberta Slip Rutile plus Cobalt Carb (Revised) · recipe #799216 ↗. It is one of the 7,063 database photos, so the app says so and leaves this search out of the session scores.



🎨 **GLAZE COLOUR**

#5A88B1 steel blue

the biggest colour area (55 % of the pot)

- R1 steel blue 55 %
- R2 cool gray 23 %
- R3 graphite black 5 %
- R4 cobalt blue 4 %

🏺 **SHAPE (most likely)**

cup 99 % sure

✨ **SURFACE (top 3)**

glossy	87 %
satin	9 %

🔎 **MORE**

transparency: **opaque** 69 % · texture: **smooth** 92 % · ceramic (CLIP) 1.00 · 2 object(s) on the photo

🧪 Top 3 recipes (photo + colour)



1

Alberta Slip Rutile plus Cobalt Carb (Revised)

View on Glazy ↗ #799216

cone 5 - 6 · Glossy

#648CAE steel blue · ΔE00 2.3

score **0.903** · family 6

photo	0.49
colour	0.41



2

Heath Ivory (Copy) testen Ile de Clay 27-02-2022

View on Glazy ↗ #210411

cone 5 1/2 - 6 · surface ?

#5F89AF steel blue · ΔE00 0.9

score **0.840** · family 6

photo	0.40
colour	0.44



3

Verano / Summer

View on Glazy ↗ #320051

cone 5 - 6 · Glossy

#3C7FBF steel blue · ΔE00 4.8

score **0.802** · family 6

photo	0.44
colour	0.36

Similar pottery (Balanced)



Alberta Slip Rutile plus Cobalt Carb (Revised)
object 799216_1 · Glazy #799216 ↗
cone 5–6 · glossy
■→■ ΔE 4.6
✓ shape ✓ glaze ✓ colour ✓ strict



Rutile Blue minspar II
object 699011_1 · Glazy #699011 ↗
cone 9–10 · surface ?
■→■ ΔE 5.6
✓ shape ✓ glaze ✓ colour ✓ strict



Kulli Okchakko
object 694803_1 · Glazy #694803 ↗
cone 5–7 · surface ?
■→■ ΔE 6.4
✓ shape ✓ glaze ✓ colour ✓ strict



Sept Floating Blue Ravenscraig
object 156908_1 · Glazy #156908 ↗
cone 5 1/2–7 · surface ?
■→■ ΔE 14.2
✓ shape ✓ glaze ✓ colour ✓ strict



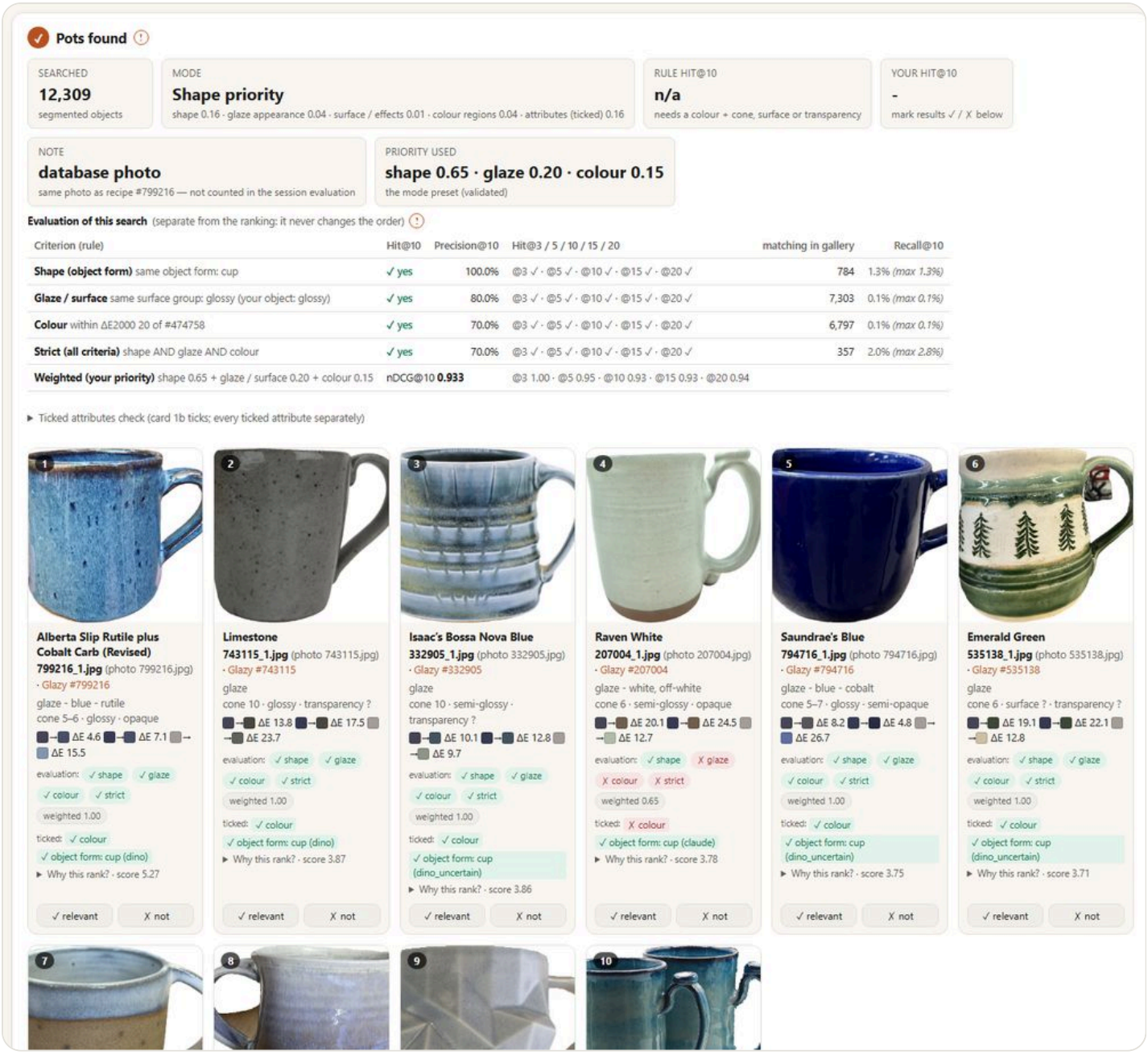
Mulberry
object 322358_1 · Glazy #322358 ↗
cone 5–6 · glossy
■→■ ΔE 7.1
✓ shape ✓ glaze ✓ colour ✓ strict



Boba Blue
object 692937_1 · Glazy #692937 ↗
cone 6–7 · glossy
■→■ ΔE 8.4
✓ shape ✓ glaze ✓ colour ✓ strict

Two colours: the blue glaze (steel blue 55 %) and dark edges (graphite black). The search uses the biggest; click another colour chip to use it instead.

Matched: its own recipe is **#1** (0.903). With the **Shape** preset (shape 0.65) the list fills with other cups; one of them, “Raven White”, is ✗ glaze ✗ colour (ΔE 20.1): a cup of the right shape in the wrong glaze. That is what Shape-first means.



Example D with the **Shape** preset: cups first, ✓ / ✗ on every pot.

6 · Example E: words only, with typos

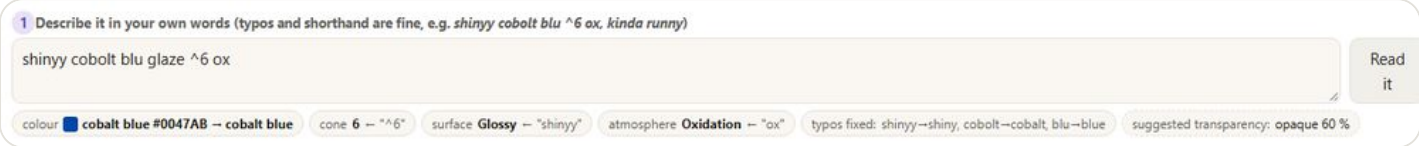
What you do

On 🧪 Glaze recipes type

`shinyy cobolt blu glaze ^6 ox` and press **Read it**, then **🔍 Find recipes**.

What the app understood

colour
cobalt blue
#0047AB · cone
6
(from "^6") · surface
Glossy
(from "shinyy") · atmosphere
Oxidation
(from "ox") · typos fixed: shinyy→shiny, cobolt→cobalt, blu→blue

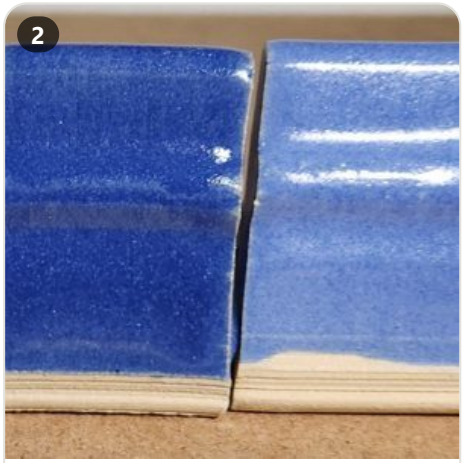




crystalline #3 blue chiaro
View on Glazy ↗ #482955
cone 6 - 7 · Glossy
#254996 cobalt blue · ΔE00 3.1
score **0.859** · family 6

words	0.30
colour	0.29
cone	0.11
surface	0.11
atmosphere	0.06

✓ cone ✓ surface



Lavender - old - full recipe (Cedar Hill)
View on Glazy ↗ #10968
cone 6 - 7 · Glossy
#2E4796 cobalt blue · ΔE00 3.8
score **0.857** · family 6

words	0.30
colour	0.28
cone	0.11
surface	0.11
atmosphere	0.06

✓ cone ✓ surface



Five1 Cobalto1
View on Glazy ↗ #462496
cone 6 - 8 · Glossy
#1F448C cobalt blue · ΔE00 4.2
score **0.848** · family 6

words	0.30
colour	0.27
cone	0.11
surface	0.11
atmosphere	0.06

✓ cone ✓ surface

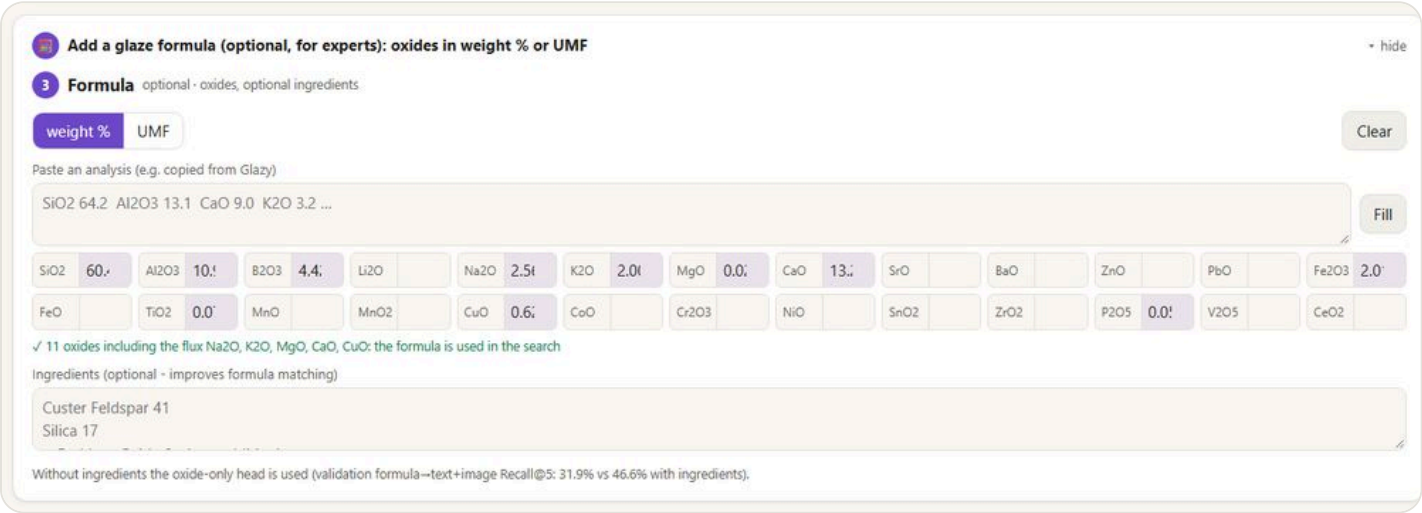
🐇 **Rabbit hole: two readers at once**

The words are used twice. (1) A **rule-based reader** finds values: colour → HEX, "^6" → cone 6, "shinyy" → Glossy, "ox" → Oxidation. Those values are scored exactly (match = 1, no match = 0). (2) A **trained text model** reads the whole sentence and compares it with every recipe. Parts here: words 0.39 · colour 0.33 · cone 0.11 · surface 0.11 · atmosphere 0.06.

On 300 noisy test sentences the top 10 had the right colour + surface + cone 54 % of the time with the text model alone, 55 % with the read values alone and **73 % with both**; that is why both are used.

7 · Example F: a chemical formula only

Open 🧮 **Add a glaze formula** on the recipes page and type (or paste) the oxides of the celadon vase's recipe, #410101, in weight %. Tick only **formula**. This recipe is in the **test split**: the smart model never saw it while training. Can the formula find it again?



SiO2	Al2O3	CaO	K2O	Na2O	Fe2O3	MgO	TiO2	B2O3	P2O5	CuO
60.4878	10.9437	13.2563	2.0088	2.5647	2.0118	0.0201	0.0739	4.4252	0.0505	0.625



Old forge first five base + 1% copper #1

View on Glazy ↗ #235893

cone 6 · surface ?

#3A332F brown black

score **0.843** · family 7

formula 0.84



SS Brushing V2 - Copper Carb (Jurassic Spek)

View on Glazy ↗ #635594

cone 7 - 8 · Glossy

#434118 dark brown

score **0.843** · family 8

formula 0.84



Southside Green Gloss

View on Glazy ↗ #736353

cone 6 · Glossy

#2F3C2F soft black

score **0.840** · family 8

formula 0.84

🐇 Rabbit hole: matched or not matched?

Its own recipe came **#11**. So with **Show 10** it is ✗ **not in the top 10** (Hit@10 = no), and with **Show 20** it is ✓ **in the top 20** (Hit@20 = yes). The top 3 are other copper + iron glazes with similar chemistry: a formula finds **similar chemistry**, not a guaranteed colour (their photos are brown-black and dark green, because firing, thickness and clay change the colour).

On all 1,059 test recipes a formula alone found the right recipe in the top 10 for 33 % of them; random guessing would give 0.9 %. A formula needs at least 3 oxides including a flux (for example CaO, K2O or Na2O); otherwise the app says so and ignores it.

8 · All answers on one page

Input	Glaze colour	Glaze recipes (top 3)	Surface (top 3)	Shape
 Glazy #614435 ↗	**#ADA869** picked on the glaze	1. Buttercream 2. Global Glaze 3. Seagull Egg	satin 39 % glossy 31 % satin-matte 15 % Glazy says: satin-matte ✗	**test tile** 100 %
 Glazy #410101 ↗	**#A4A992** sage green	1. Jeannine's Sagebrush (SAC) 2. Yoko's Leaf Green 3. Vert Crackle (ES)	satin 42 % matte 24 % satin-matte 16 % Glazy says: satin ✓	**vase** 95 %
 Glazy #577829 ↗	**#1E345B** navy blue	1. Storm Blue - Blue Room Ingredients 2. MG2 1Co 1Cu 3. 12 - Blue haze based on kitten's clear adapt with Gillespie	glossy 95 % Glazy says: glossy ✓	**bowl** 97 %
 Glazy #799216 ↗	**#5A88B1** steel blue	1. Alberta Slip Rutile plus Cobalt Carb (Revised) 2. Heath Ivory (Copy) testen Ile de Clay 27-02-2022 3. Verano / Summer	glossy 87 % satin 9 % Glazy says: glossy ✓	**cup** 99 %
`“shinyy cobolt blu glaze ^6 ox”`	**#0047AB** cobalt blue	1. crystalline #3 blue chiaro 2. Lavender - old - full recipe (Cedar Hill) 3. Five1 Cobalto1	Glossy (from “shinyy”)	-
`formula of #410101`	-	1. Old forge first five base + 1% copper #1 2. SS Brushing V2 - Copper Carb (Jurassic Spek) 3. Southside Green Gloss its own recipe: rank 11	-	-

Glaze colour = the picked colour (A) or the biggest colour area (B–D). Surfaces and shapes are model probabilities. “Glazy says” is the surface written on the recipe's own Glazy page.

9 · What the numbers mean

Number	In simple words	Exactly

% on My photo	How sure the model is	The trained probability of that value (top 3 shown). Not a measured accuracy.
ceramic (CLIP)	Is it pottery?	CLIP's ceramic score, 0–1; objects below 0.35 are not selected by default.
score	How close a recipe is	Weighted sum of the parts below, 0–1. Only the parts you gave count; their weights are scaled to add up to 1.
ΔE00	How different two colours look	CIEDE2000 colour difference (made to follow human eyes). 0 = the same.
colour part	Colour closeness	exp(−ΔE00 / 20), see the table below.
words / photo / formula part	How well the recipe fits	(cosine similarity + 1) / 2 in the smart model's shared space.
learned match	Words ↔ chemistry	Colour match model: (cosine + 1) / 2 between the words and the recipe's chemistry.
cone · surface	Right firing? right finish?	1 when it matches (cone ranges overlap; 04 ≠ 4), else 0.
family	A chemistry group	K-Means group. *Explore formulations* takes at most 2 per family, to show different recipes for a similar look.
✓ / ✗ / ?	Matches or not	Shape: same form. Glaze: same surface group (glossy / satin / matte). Colour: ΔE ≤ 20. Strict: all three. ? = no label.
Hit@K	Any good one near the top?	Yes if at least one of the first K results matches.
P@K	How many good ones?	Matching results among the first K, divided by K.
Recall@K	Found all of them?	Share of all matching items that reach the top K. Shown for the true recipe ID, and in the model tests (Recall@10 = the right recipe in the top 10).
nDCG@K	Best ones first?	0–1; 1 = the best possible order (graded by your priority).
Hit Rate@K	Over many searches	Share of searches with Hit@K = yes (📊 Scores page).

ΔE00	colour part	what it looks like
0	1.00	the same colour
2	0.90	hard to see the difference
5	0.78	a small difference side by side
10	0.61	a clearly different shade
20	0.37	a different colour (the limit of a colour match in the scores)

10 · Photo + words + formula together

Yes, you can give all three in one search. On 🧪 Glaze recipes tick what to use (text, colour, photo object, formula). Each part has a weight: words 0.35, photo 0.35, formula 0.25, colour 0.30, cone 0.10, surface 0.10, transparency 0.05, atmosphere 0.05. **Only the parts you give count**, and their weights are divided by their sum, so they always add up to 1:

You give	Weights used
only a formula (example F)	formula 1.00
photo + colour (example A, step 2)	photo 0.54 · colour 0.46
words with colour, cone, surface, atmosphere (example E)	words 0.39 · colour 0.33 · cone 0.11 · surface 0.11 · atmosphere 0.06
photo + colour + words with cone and surface (example A, step 3)	words 0.29 · photo 0.29 · colour 0.25 · cone 0.08 · surface 0.08

So one input alone gets all the weight, and several inputs share it in fixed proportions. On 🏺 Similar pottery the same idea holds: photo 0.40, words 0.35, formula 0.25 when given; the sliders split the photo part between shape, glaze and colour.

11 · Tips and problems

If…	Then…
the colour is the bare clay or a shadow	On 📷 My photo switch to 🎨 **Pick colour** and click on the glaze; or click another colour chip in **What the app found**.
the wrong pot is used	Click the pot you mean (yellow outline = the one in use).
the surface guess looks wrong	It often is (45 % right first): type the surface (e.g. *satin*) in the recipe words, or choose it in **Surface**.
results ignore your cone	Choose it in **Firing cone**; remember cone 04 (low fire) is not cone 4 (mid fire).
no recipe photos	They come from Glazy: check the internet. The search itself still works.
“the formula is not used”	Give at least 3 oxides including a flux (CaO, K2O, Na2O, MgO, ZnO, …).
the page says the server is not running	Start RUN_GLAZE_SEARCH.bat again and keep its window open.
a recipe looks perfect	Open it with **View on Glazy** and read it there; test-fire a small tile before you mix a big batch.

12 · Where things are

What	Where

This manual in the app	The 📖 **How to use** button (and **Manual** at the top right), or http://127.0.0.1:8770/manual/CG2_Glaze_Search_Manual.html while the app runs
This manual as files	docs\manuals\CG2_Glaze_Search_Manual.html (open it in any browser, works offline) and CG2_Glaze_Search_Manual.pdf
Your uploads and searches	multimodal_experiments\work\queries\ (session_log.json for pottery searches, glaze_search_log.jsonl for recipe searches)
Technical details	docs\glaze_integration_architecture.md, docs\SEARCH_GUIDE.md